V2X and Deep Reinforcement Learning-Aided Mobility-Aware Lane Changing for Emergency Vehicle Preemption in Connected Autonomous Transport Systems

Cao Ding, Ivan Wang-Hei Ho, *Senior Member, IEEE*, Edward Chung, and Tingting Fan

***Abstract*— Emergency vehicle preemption (EVP) aims to provide the right-of-way to emergency vehicles (EVs) so that they can travel to the incident location efficiently. The travel time of EVs is the most important indicator of EVP efficiency, which should be minimized by distinct methods or algorithms. However, conventional EVP methods using strobe emitters, light emitters or sirens performs poorly in high-density vehicular traffic. Vehicle-to-everything (V2X) communication plays a pivotal role in intelligent transportation systems (ITS), which can assist EVs to travel safely and efficiently in connected autonomous transport systems (CATS). Enabled by V2X, this paper proposes a deep reinforcement learning-aided mobility-aware lane change algorithm (DRL-MLC) to enhance the efficiency of EVP. In the first stage, the EV learns to change lane based on a policy-based deep reinforcement learning (DRL) algorithm to find the shortest trajectory. In the second stage, autonomous vehicles (AVs) perform mobility-aware lane changing (MLC) to make way for the EV based on the emergency messages (EM) they received. Note that the performance of DRL-MLC strongly relies on the quality of service (QoS) of V2X, and improper network parameters of the on-board units (OBUs) that do not match with the vehicular density will significantly degrade the QoS. Therefore, in the third stage, the proposed algorithm fine-tunes specific parameters including communication range, carrier sensing range, packet rate, and contention window according to the real-time vehicular density based on a curve-fitting optimization method. Our results indicate that at medium-to-high density (e.g., 0.15 veh/m), the average speed of DRL-MLC has more than 49% average improvement than traditional lane changing models, and the ten-minute target travel time for EVs can be achieved by 95% with the proposed algorithm.**

***Index Terms*— V2X, vehicular networks, lane change, emergency vehicle preemption.**

I. Introduction

THE frequency of emergency calls is increasing as the number of automobiles, accidents, and disasters in cities continue to climb. As a consequence, emergency vehicles (EVs) such as ambulances, fire trucks, and police cars are anticipated to arrive at accident sites as quickly as possible to assist with the rescue. Therefore, numerous researchers have proposed various intelligent transportation system (ITS) applications [1] for emergency vehicle preemption (EVP), which enable EVs to arrive at accident sites in the shortest possible time. Traditional EVP technologies like light emitters, strobe emitters, and alarm detectors have drawbacks including blocked line-of-sight (LoS) and excessive noises [2]. ITS-enabled EVP applications, on the other hand, have better safety, efficiency, and reliability [3].

Vehicular networks that facilitate drivers or autonomous vehicles (AVs) are enabled by dedicated short-range communication (DSRC) [4] or cellular vehicle to everything (C-V2X) [5] protocols, which play a critical role in ITS applications for enhancing traffic efficiency and road safety. There is a large body of work on exploiting V2X for EVP. Unibaso et al. [6] proposed a traffic signal preemption algorithm that takes into account vehicle spatial locations and driving angles in cooperative awareness messaging. Chen et al. [7] demonstrated ITS firmware for EVP that is based on the National Transportation Communications for ITS Protocol (NTCIP). They were able to achieve V2X communications and data interoperability between automobiles and roadside equipment using NTCIP. Previous studies, on the other hand, ignored the effect of mobility on the dissemination of emergency messages (EMs), assuming a static distribution of vehicles on the road with no hidden terminals [8]. However, in reality, vehicular mobility is controlled by several factors such as vehicular density, road topology, and driving strategy. These factors can cause delays and interruptions in the transmission of EMs. Although several mobility research on the dissemination of EMs have been conducted [9], there is a lack of empirical studies on the impact of EMs on EVP.

To fill the gap, this paper proposes a V2X-aided deep reinforcement learning mobility-aware lane change algorithm (DRL-MLC) for EVP in connected autonomous transport systems (CATS), where vehicles are equipped with sensors, global positioning systems (GPS), and on-board units (OBUs) to collect or broadcast information such as position from or to other vehicles within the communication range. The usage of roadside units (RSUs) for broadcasting emergency messages is also considered in this paper. The novelty of this paper

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Cao Ding, Edward Chung, and Tingting Fan are with the Department of Electrical and Electronic Engineering, The Hong Kong Polytechnic University, Hong Kong (e-mail: caosnow.ding@connect.polyu.hk; edward.cs.chung@polyu.edu.hk; tingfan@polyu.edu.hk).

Ivan Wang-Hei Ho is with the Department of Electrical and Electronic Engineering, The Hong Kong Polytechnic University, Hong Kong, and also with the Otto Poon Charitable Foundation Smart Cities Research Institute, Hong Kong (e-mail: ivanwh.ho@polyu.edu.hk).

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is the combination of V2X environment and lane changing control of the EV and AVs. Such an approach considers the impact of V2X and control algorithms on cooperative autonomous driving. Overall, the main contributions of this work are three-fold:

i) Based on the proximal policy optimization (PPO) deep reinforcement learning (DRL), we propose an EV lane change model in the DRL-MLC, in which the EV learns how to change lanes at the appropriate time and position and find the shortest trajectory. The proposed algorithm significantly reduces the travel time while guaranteeing safety and comfort. At medium density, the DRL-MLC improves the EV's average speed performance by more than 81.8% and 41.3%, respectively, over the lane changing model in the simulation of urban mobility (SUMO-LC) [10] and support vector machines-based lane changing model (SVM-LC) [11].

ii) In addition, this work also considered the mobility-aware lane changing (MLC) algorithm for AVs to make way for the EV. With the MLC, the blocking AVs will change to the optimal adjacent lane to reduce the blocking time and further optimize the travel time of the EV. The selection of the optimal adjacent lane tends to be the lane that currently has enough space for LC and will not cause flow breakdown.

iii) The quality of service (QoS) of V2X messaging is also considered in this paper, and the QoS-V2X algorithm is proposed to improve network performance and reliability by updating specific parameters based on real-time vehicular density. After implementing QoS-V2X, DRL-MLC effectively reduces emergency message latency and enhance packet reception rate (PRR). The EM delay of DRL-MLC is reduced by 59.3% at a relatively high density (0.2 veh/m) compared to DSRC, which does not use a mobility adaptation algorithm. At a relatively low density (0.15 veh/m), the PRR of DRL-MLC is 41% higher than that of DSRC.

The structure of the paper is as follows. Section II introduces the related work. Section III introduces the system models. Section IV presents the DRL-MLC algorithm. Section V gives the numerical results of the proposed algorithms and multiple benchmarks. Finally, Section VI concludes the paper.

II. Related Work

To the best of our knowledge, there are few lane change (LC) models tailored for EVP scenarios in CATS. However, there are a large number of LC models for the ego vehicle that can be substituted for the EV. There are two major differences between the LC model for the EV and the general LC models. First, the EV has a higher priority than other vehicles, whereas in general LC models the ego vehicle tends to have the same priority as other vehicles. Second, the reaction of other vehicles to the EV, such as giving way and slowing down. In general LC models, other vehicles do not respond to the ego vehicle, they maintain the original driving state during the ego vehicle LC process. Therefore, the difficulty of EV lane changing is to ensure the priority of EV and take into account the reaction of other vehicles in the decision-making process.

The general LC models can be divided into two categories according to their working principles. The first category is the rule-based models, which has a large number of predefined rules and conditions and selects a certain rule to execute according to the current condition of the EV. For instance, in [10], the ego vehicle in the SUMO-LC model decides the moment of LC and the target lane as well as the speed adjustment based on the urgency, distance from the end of the lane, density of each lane, spacing of neighboring cars, etc. The second category is the learning-based models, which use machine learning to optimize lane-changing decisions and interact with the environment through a trial-and-error mechanism. Liu et al. [11] proposed an SVM-LC model, which considers the benefits and safety of lane changing, and uses Gaussian kernel SVM to solve the multi-parametric and nonlinear problem of LC decision-making process. In [12], a random forest-based lane change (RF-LC) model was proposed, which extracted 1108 sample data of lane changing or lane keeping for two road segments in the next generation simulation (NGSIM) dataset for training the random forest decision-making model and 277 sample data for testing. The result shows that the accuracy of RF-LC for LC prediction of AVs is 97.6% at a decision tree number of 700. The DRL-based LC models are currently the most advanced learning-based models. It is mainly divided into two types to improve LC performance. The first type is to optimize the state, action, and reward space in Markov decision process (MDP). In [13], the Aerial View image vector is used as a better state input to the neural network. The second type is to optimize the DRL framework and LC models. For example, Zhang et al. [14] proposed a multi-agent DRL framework to train LC models with collaboration awareness.

The deficiencies of these LC models are obvious. The first category cannot make proper decision beyond the predefined rules and conditions, which is not applicable in the real world with changing traffic conditions. The second category is limited by the training data, and they need to be supported by a large amount of LC data from different scenarios. Specifically, DRL-based LC models focus on the ego vehicle, and ignore the interaction between vehicles. Moreover, these LC models never consider the impact of V2X messages and interactions among connected vehicles. In this paper, DRL-based LC and V2X messages are jointly considered in the decision-making models. V2X messages affect the lane changes of AVs, so that the DRL environment changes in real time. The EV makes LC decisions based on the dynamic environment taking into account the interaction with other connected vehicles via V2X.

In addition to LC control, the QoS of V2X is also a critical part of EVP. Such a vehicular channel is limited by the capacity and interference from neighboring vehicles, and this will affect the LC performance. Current research on QoS focuses on tuning controllable communication parameters (e.g., data rate and communication range [15]) to achieve optimal QoS. Sommer et al. [16] proposed an adaptive traffic beacon (ATB) message dissemination algorithm. In this algorithm, the beacon interval is calculated and adjusted according to the channel quality and message utility, and beacons are sent as frequently as possible while guaranteeing a congestion-free channel to

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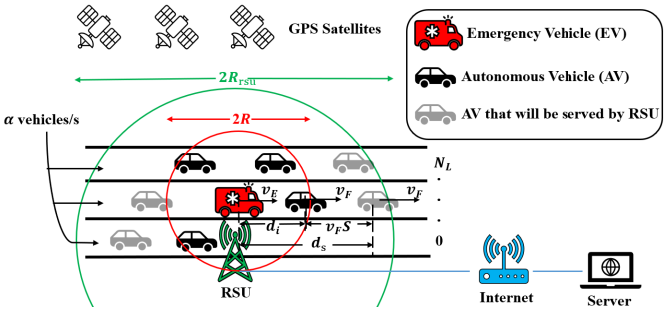

Fig. 1. The connected autonomous vehicle environment.

exchange information in the knowledge base. In [17], an adaptive and mobility-based algorithm (AMBA) was proposed. Since AMBA not only adjusts the beacon interval but also takes into account the control of communication and carrier sense ranges based on the average speed of vehicles, the results show that the transmission delay and success rate of AMBA is better than that of ATB. In [18], a multihop emergency message broadcast (MEMB) scheme was proposed. MEMB establishes the priority of EMs, where emergency warnings are divided into three levels, and each level adopts different broadcast settings. The results show that such an event-driven broadcast scheme can reduce transmission delay and ensure reliability.

In our work, SUMO-LC, SVM-LC, RF-LC, ATB, AMBA, and MEMB algorithms are reproduced and customized according to our system model. They serve as benchmarks for peer comparison with our proposed algorithm.

III. System Models

As illustrated in Fig. 1, we consider a 1-dimensional V2X communications-aided EVP scenario. The EV serves as the transmitter, sending EMs to moving AVs within its communication range. RSUs are deployed to relay the EM to vehicles within an extended area.

A. Mobility Model

The origin and destination of the EV and AVs are arbitrarily defined, and the arrival of vehicles at each lane is modeled as a poisson process [19] with α_i vehicles/s for the i-th lane, so the total number of vehicles that arrives at the road per second is $\alpha = \sum_{i=0}^{N_L} \alpha_i$. Note that N_L represents the total number of lanes on the road. The speed, acceleration, and braking behavior of each vehicle will be regulated by the safety gap control in the car-following model between the lowest velocity (v_{min}) and the road's speed limit (v_{max}). To ensure minimal headway between vehicles on the road, the car-following models for the EV and AVs use numerous control modes (e.g., collision avoidance control mode, gap control mode, etc.). Assume that the current speed of the following EV is v_E and the current speed of the following AV is v_F, the safety distance d_s from the preceding vehicle can be expressed as

$$d_s > \begin{cases} v_E t_s, & \text{for EV}, \\ v_F t_s, & \text{for AV}, \end{cases} \tag{1}$$

where t_s is the safety time, which is the response time for a driver to react to an event. The minimum safety distance d_s is a variable and depends on the assumed t_s and the speed of the following vehicle.

B. Vehicular Fading Channel

Since buildings and vehicles can reflect, diffract, or even block signals in V2X communications, the overall received signal consists of reflected signals with randomly distributed amplitudes and phases. To accurately model the vehicular network propagation channel, some fading models are proposed and widely applied, such as the Rayleigh fading model [20], Nakagami model [21], etc. The probability density function (pdf) of the received signal power γ is

$$P(\gamma) = (\frac{\iota}{P_r})^{\iota} \frac{\gamma^{\iota-1}}{\Gamma(\iota)} e^{-\frac{\iota\gamma}{P_r}}, \quad \gamma \geqq 0 \tag{2}$$

where $\Gamma(\iota)$ is the Gamma function of the Nakagami fading factor ι. For $\iota = \frac{(\Upsilon+1)^2}{2\Upsilon+1}$, the fading approaches a Rician distribution. Υ is the ratio of the line-of-sight (LoS) power to the non-line-of-sight (NLoS). For $\iota = 1$, the fading reduces to Rayleigh distribution. $P_r = \frac{P_t K}{x^q}$ is the average received power, where P_t is the transmission power, $K = G_t G_r (c/4\pi f)^2$, in which c is the speed of light, G_t and G_r are the antenna gains of the transmitter and receiver, and $f = 5.9$ GHz is the carrier frequency. x is the distance in meters, and q is the path-loss exponent.

C. Emergency Message Delay

When an EV encounters an emergency mission, it will broadcast emergency packets and designate relay nodes to rebroadcast the message to more neighbors. The relay node is usually the farthest node within the communication range, and the EM continues to propagate until it reaches a distance L determined by the message itself. Since RSUs have a significantly wider communication range than vehicles, allowing a larger number of vehicles to receive the rebroadcasted message at the same time, they are chosen as relay nodes in this work. EMs will be transmitted using high-priority access category 3 (AC3) after sensing an idle channel for an AIFS$\cdot\varrho$ duration. Where AIFS is the arbitration inter-frame spacing in EDCA before the commencement of a transmission, $\cdot$ is a product, and $\varrho = 13$ μs is the length of a time slot [22]. If there is no idle channel, the AC0 will select a contention window from the range [0, W_c], where $W_c = 3$ in the setting, and the back-off counter will begin. At the MAC layer, the average latency for an emergency packet to be ready for transmission is represented as

$$T_{mac} = \sum_{i=0}^{W_c-1} \frac{p}{W_c} \sum_{k=0}^{i} (pt_p) + t_p, \tag{3}$$

$$= \frac{p^2 t_p (W_c - 1)}{2} + t_p, \tag{4}$$

where $p \in [0, 1]$ is the conditional busy channel probability. The relay must rebroadcast the EM to the next hop after receiving it. The network topology persistence protocol (NTPP) [23]

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TABLE I
V2X MESSAGING PROTOCOLS

Index	Transmitter	Receiver	Message Content	Receiver's Reaction
1	EV	AV	Msg 1	Ret 1
2	EV	RSU	Msg 2	Ret 2
3	AV	RSU	Msg 3	Ret 3
4	RSU	EV	Msg 4	Ret 4
5	RSU	AV	Msg 5	Ret 5

and Greedy Perimeter Stateless Routing (GPSR) [24] are two existing techniques for choosing the best relay based on the shortest transmission path. In this study, the farthest RSU from the sender will have a high rebroadcasting probability P_{rt} and a short waiting time T_w as

$$P_{rt}(l) = \frac{(\frac{l}{R_{rsu}}) + (1 - \frac{\rho_c}{\rho_{max}})}{2}, \tag{5}$$

$$T_w(l) = (1 - \frac{l}{R_{rsu}}) \cdot (\frac{\rho_c}{\rho_{max}}) \cdot (2t_p + \delta), \tag{6}$$

where l is the distance between the sender and the RSU relay, ρ_c is the current vehicle density, ρ_{max} is the maximum vehicle density, and δ is the propagation delay. The average emergency message time delay to reach a distance L_k is

$$T_{EM} = \left\lfloor \frac{L_k}{l} \right\rfloor (T_{mac} + \frac{T_w(l)}{P_{rt}(l)}). \tag{7}$$

D. V2X Messaging Protocols

Our system assumes that all vehicle nodes on the road are connected, that is, EV, AV, and RSU transmit and receive messages from each other at the same interval. As shown in Table I, V V2X application layer messages are designed in detail as protocols, which are divided into five types of messages according to different transceivers. In the V2X messaging protocols, in addition to specific headers and addresses, we design distinct message content to realize specific algorithms, or the receiver's reaction, such as AV mobility-aware lane change (MLC). The definitions of the specific message content and the receiver's reaction are as follows:

- Msg 1, Msg 2, and Msg 5 have the same message content, which includes the current road and lane of the EV (R_E,L_E), the lane change state of the EV ($state_{LC}$), and the lateral position, longitudinal position of the EV.
- Ret 1 is the same as Ret 5, that is, the MLC algorithm is enabled to make the AV give way to the rear EV. Ret 2 is the RSU rebroadcasts the received Msg 2 in a larger range, so that more AVs can change lanes to give way in advance.
- Msg 3 contains the state of the AV (i.e., lateral position, longitudinal velocity, longitudinal acceleration, longtitudinal distance between the AV and the EV). Msg 4 contains the states of potential AV neighbors from the knowledge base.
- Ret 3 is the RSU gain centralized knowledge of its local area and save the states of potential AV neighbors to the EV in Msg 4. Ret 4 is the EV gathers the states of itself and AV neighbors to perform DRL-MLC algorithm.

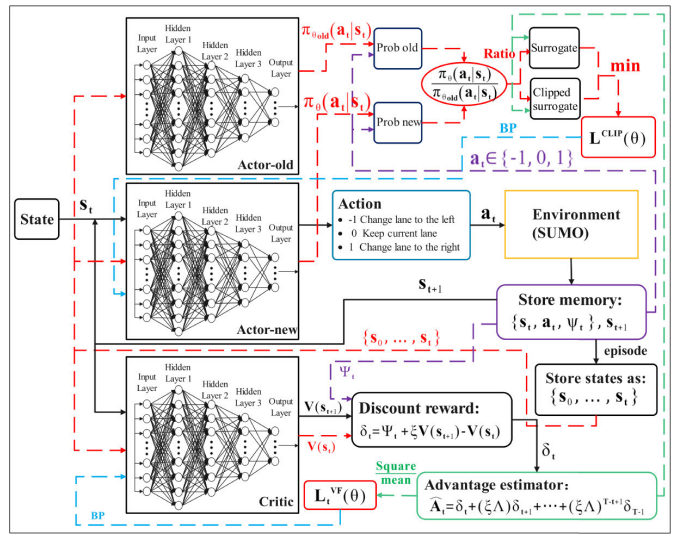

Fig. 2. The PPO-based DRL architecture.

In summary, V2X messaging protocol 1, 2, and 5 are communication processes that implement the MLC algorithm of AV. V2X messaging protocol 3 and 4 provide the states of neighboring AVs to the DRL algorithm for training.

IV. V2X-AIDED DEEP REINFORCEMENT LEARNING MOBILITY-AWARE LANE CHANGING ALGORITHM

In this chapter, we present the proposed algorithms in a step-by-step manner. In the first stage, the EV learns to change lanes at the right time and location through the DRL algorithm, so as to find the shortest trajectory from the starting point to the destination. In the second stage, we customize the MLC algorithm for AVs, where AVs in front of the EV change lanes to give the right of the way. Note that the performance of both DRL and MLC is highly dependent on the quality of service (QoS) of V2X messages, which is deeply affected by the vehicular density on the road. Therefore, in the third stage, we fine-tune the network parameters according to the current vehicular density to enhance the QoS, so as to improve the performance of the DRL and MLC algorithms as much as possible.

A. DRL-Based Lane Change for the EV

General LC models (e.g., SUMO-LC) provide acceptable average speeds for the EV, but they only consider AVs in a small range in front of the EV for lane-changing strategies, and their performance is limited. Therefore, we propose a policy-based DRL to train the EV to make suitable lane changes in the V2X environment based on EV's and surrounding AV's states, as shown in Fig. 2. The loss function for the policy gradient estimator has the general formula [25] as

$$L^{PG}(\theta) = \hat{\mathbb{E}}_t[\log \pi_\theta(a_t|s_t)\hat{A}_t], \tag{8}$$

where θ is the parameter vector, $\hat{\mathbb{E}}_t$ is the expectation operator, π_θ is a random policy, a_t and s_t represent the action and state at time step t, respectively. $\hat{A}_t$ is the advantage function estimator. Some sophisticated algorithms, such as proximal

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policy optimization (PPO) [25], employs an actor-critic (A2C) approach, which combines the benefits of value-based and policy-based methodologies. Since the PPO algorithm is simple to implement and has low computational complexity and burden, it is used in this study to implement the DRL. The loss function in PPO is made up of a policy clipped surrogate, a value function error, and an entropy bonus, which can be expressed as

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t[min(r_t(\theta)\hat{A}_t, clip(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)], \quad (9)$$

$$L_t^{CLIP+VF+S}(\theta) = \hat{\mathbb{E}}_t[L_t^{CLIP}(\theta) - c_1 L_t^{VF}(\theta) + c_2 S[\pi_\theta](s_t)], \quad (10)$$

where $L^{CLIP}(\theta)$ is the loss function of the policy clipped surrogate, $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ represents the probability rate, $\epsilon = 0.2$ is the hyperparameter, $clip(r_t(\theta), 1-\epsilon, 1+\epsilon)$ revises the surrogate by clipping the probability ratio; $L_t^{CLIP+VF+S}(\theta)$ is the total loss function, c_1 and c_2 are coefficients, $L_t^{VF}(\theta) = (V_\theta(s_t) - V_t^{targ})^2$ is the error of value function, and $S[\pi_\theta](s_t)$ denotes the entropy bonus. In each iteration, the PPO methodology utilizes a fixed-length segment, say T-trajectory, which implies that the algorithm runs the policy for t ∈ [0,T] time steps while each parallel actor collects data. The abbreviated version of the advantage estimator is as follows

$$\hat{A}_t = \delta_t + (\xi\Lambda)\delta_{t+1} + \cdots + (\xi\Lambda)^{T-t+1}\delta_{T-1}, \quad (11)$$

where $\delta_t = \Psi_t + \xi V(s_{t+1}) - V(s_t)$, Ψ_t is the total reward, and ξ is the discount factor. As shown in Fig. 2, the specific operations in each iteration are as follows.

- The total state $\boldsymbol{s_t}$ is used as the input to Actor-new neural network to obtain the action a_t, which generates the next state $\boldsymbol{s_{t+1}}$ and reward Ψ_t through the environment (SUMO), and stores them in memory. Note that the weights in Actor-new are not updated at this point.
- The state combinations $\{\boldsymbol{s_0}, \ldots, \boldsymbol{s_t}\}$ and $\boldsymbol{s_{t+1}}$ are used as the input to the Critic network to obtain the corresponding value functions $V(s_t)$ and $V(s_{t+1})$ respectively. $V(s_t)$ and $V(s_{t+1})$ are used to calculate the discount reward δ_t, and then estimate the advantage $\hat{A}_t$. After squaring $\hat{A}_t$ and dividing the average value, the value function loss $L_t^{VF}(\theta)$ is obtained, and $L_t^{VF}(\theta)$ is used to update the weights of the Critic network via backpropagation (BP).
- Finally, state combinations $\{\boldsymbol{s_0}, \ldots, \boldsymbol{s_t}\}$ are used as input to Actor-new and Actor-old to get the corresponding action combinations $\pi_\theta(a_t|s_t)$ and $\pi_{\theta_{old}}(a_t|s_t)$ respectively, and divide them to get the probability rate $r_t(\theta)$, which is then used together with $\hat{A}_t$ to calculate the surrogate and clipped surrogate. Actor-old network then take the minimum value of the clipped surrogate, $L^{CLIP}(\theta)$, to update Actor-new via BP and update Actor-new's weights to Actor-old.

Through this on-policy approach, the EV can continuously update the Actor-new network in the SUMO environment in real time to optimize the lane change policy. The specific settings of states, actions, and rewards in this DRL algorithm are shown as follows.

1) State Space: We consider the states of the EV and potential AV neighbors involved in the lane change decision as state space representation. AVs periodically broadcast beacons (i.e., Msg 3) to the RSU and its neighbors within communication range, and the RSU is assumed to gain centralized knowledge of its local area and to send the states of potential AV neighbors to EVs. We gather four features to describe the state (e.g., $\boldsymbol{SA_j(t)}$) of time t, which are lateral position ($pl_j(t)$), longitudinal velocity ($v_j(t)$), longitudinal acceleration ($ac_j(t)$), and longitudinal distance ($d_j(t)$) between the AV and the EV, for the i-th AV within the communication range of RSU (R_{rsu}). Five features are collected for the agents (i.e., EVs) to represent the state (i.e., $\boldsymbol{SE_i(t)}$), which has the same features as AV's but with the longitudinal position ($po_i(t)$) added. Note that the velocity and position in the features are absolute, only the longitudinal distance is relative. The state space of time t can be expressed as

$$\boldsymbol{SA_j(t)} = \{pl_j(t), v_j(t), ac_j(t), d_j(t)\}, \quad (12)$$

$$\boldsymbol{SE_i(t)} = \{pl_i(t), v_i(t), ac_i(t), d_i(t), po_i(t)\}, \quad (13)$$

$$\boldsymbol{s_t} = \{\boldsymbol{SA_1(t)}, \ldots, \boldsymbol{SA_j(t)}, \boldsymbol{SE_1(t)}, \ldots, \boldsymbol{SE_i(t)}\}, \quad (14)$$

In simulation, we consider the case of up to one EV and six potential AV neighbors. Therefore, a total of 29 vehicle features are put together to generate the whole state (i.e., $\boldsymbol{s_t}$).

2) Action Space: The action space a_t in this study only considers the lateral direction so that the agent can learn how to appropriately perform lane changes in the V2X environment. There are three possible actions (i.e., -1, 0, 1); '-1' signifies lane changing to the left, '0' represents staying in the current lane without lane changing, and '1' represents lane changing to the right.

3) Reward Function: For the design of the reward function, the travel efficiency of the EV is the first tendency, followed by driving safety and comfort. The following is the design of the reward function:

(1) The speed reward analyzes three factors: the difference between the EV speed V_E and the desired speed $V_{desired}$, the difference between the EV positions after LC (P_E) and before LC (P_E'), and the time spent on LC (ϵt). Note that $\epsilon = 3600$ is a constant, which converts the collected time unit from hours to seconds. These three factors' rewards, as well as the total reward Ψ_V, can be stated as

$$\begin{cases} \Psi_{Speed}(t) = V_E - V_{desired}, \\ \Psi_{Position}(t) = -|P_E - P_E'|, \\ \Psi_{Time}(t) = -\epsilon t, \end{cases} \quad (15)$$

$$\Psi_V(t) = w_{sf} \cdot \Psi_{Speed}(t) + w_{pf} \cdot \Psi_{Position}(t) + w_{tf} \cdot \Psi_{Time}(t), \quad (16)$$

where w_{sf}, w_{pf}, w_{tf} are the weights of the speed factor, position factor, and time consumption factor respectively. Ψ_{Speed}, $\Psi_{Position}$, and Ψ_{Time} are their corresponding rewards.

(2) Vehicle crashes during lane changes, including collisions between AVs or between EV and AV are taken into account by the safety reward Ψ_S. If a collision occurs after the EV chooses the lane change strategy in action space, each collision will be

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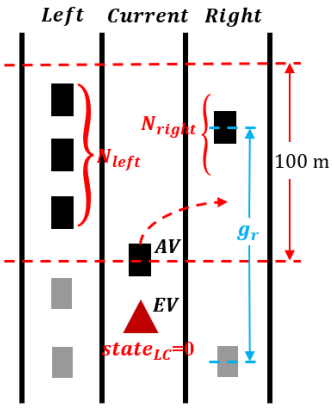

Fig. 3. Vehicular density calculation.

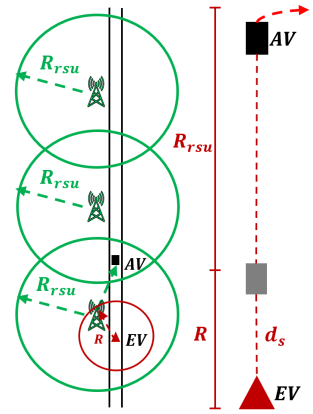

Fig. 4. MLC operating distance.

severely penalized; however, if no collision occurs, each action will get a small reward. The safety reward can be stated as

$$\Psi_S(t) = \begin{cases} +5, N_{co} = 0, \\ -100 \cdot N_{co}, N_{co} > 0, \end{cases} \tag{17}$$

where N_{co} is the number of collisions that occurred during lane changing. Since the penalty for collision is severe, the EV will do everything to avoid colliding during training.

(3) The comfort reward Ψ_C considers the driving experience and determines the amount of driving comfort by computing the jerk (i.e., the lateral and longitudinal acceleration changes). If the jerk is considerable, it indicates that the vehicle is experiencing an abrupt shift in acceleration while driving, which is unpleasant. The comfort reward can be expressed as

$$\Psi_C(t) = -[m_x \cdot \dot{\alpha_x}(t)^2 + m_y \cdot \dot{\alpha_y}(t)^2], \tag{18}$$

where m_x, m_y are the weights of the lateral and longitudinal acceleration, $\dot{\alpha_x}$ is the lateral acceleration, and $\dot{\alpha_y}$ is the longitudinal acceleration. These three rewards will be eventually added together to be the total reward Ψ_t according to the predefined weights W_V, W_S, and W_C as follows:

$$\Psi_t = W_V \cdot \Psi_V(t) + W_S \cdot \Psi_S(t) + W_C \cdot \Psi_C(t). \tag{19}$$

We normalize each reward function or factor (e.g., Ψ_V, Ψ_{Speed}) so that they are all on the same scale (i.e., [−1,1] interval). In addition, the weights of reward functions or factors (e.g., W_V, W_S, W_C) are limited to [0,1] and sums to 1. Such settings can eliminate the influence of different sizes and units on the reward function.

To summarize the DRL algorithm, the agent takes the action that will get the EV to its destination as quickly as possible in a safe and comfortable manner.

B. Mobility-Aware Lane Change (MLC) for AVs

In order to let the EV pass quickly, this paper proposes the MLC algorithm to make the blocking AVs give way to the EV by changing lanes to a lane that has greater opportunity for lane changing. The MLC algorithm is shown in Fig. 5. First of all, MLC works only when AVs receive EMs (i.e., Msg 1, Msg 5) transmitted by the EV or RSU. As shown in Fig. 3, $state_{LC}$ is to ensure the priority of EV lane change. Only when the EV is not changing lane (i.e., $state_{LC} = 0$), the blocking AV will take MLC. There is a special case where the AV is in the rightmost lane or the leftmost lane of the road, the AV has no choice but to change into another side. In normal cases, the algorithm prioritizes the adjacent lanes with sufficient gap (i.e., g_{critic}) for lane change. We adopted the mean of the sufficient gaps for lane changes with respect to empirical data [26], that is, $g_{critic} = 4.04$ seconds. Assuming that the right lane is the target lane, the gap can be calculated as

$$g_r = T_2 - T_1, \tag{20}$$

where the time (T_1) and position of the leading vehicle in the target lane were recorded at the moment the AV made lane change decision. The time (T_2) was noted when the following vehicle crossed the same position. The time difference is the gap for lane change.

If sufficient gaps are available or not available for both adjacent lanes, it will select the lane with the lower density. The blocking AV calculates the vehicular density within 100 meters ahead of itself, and obtains the vehicular densities of the left and right lanes (i.e., ρ_l and ρ_r) as follows

$$\begin{cases} \rho_l = \dfrac{N_{left}}{100}, \\ \rho_r = \dfrac{N_{right}}{100}, \end{cases} \tag{21}$$

where N_{left} and N_{right} are the number of vehicles detected on the left and right lanes. We analyze the operating distance of the MLC algorithm, that is, the distance from the transmitter to the AV making lane change, as illustrated in Fig. 4. It is assumed that the placement of RSUs covers the entire road network with a 20% overlapping area. The shortest operating distance of MLC is the safe distance between the AV and the EV (i.e., d_s). The farthest operating distance is the communication radius of the EV plus the communication radius of the RSU (i.e., $R + R_{rsu}$). Therefore, to ensure the availability of the MLC algorithm in the whole road network, it is necessary to set the appropriate communication radius of EV and RSU. The requirement is $R_{rsu} \gg R > d_s$.

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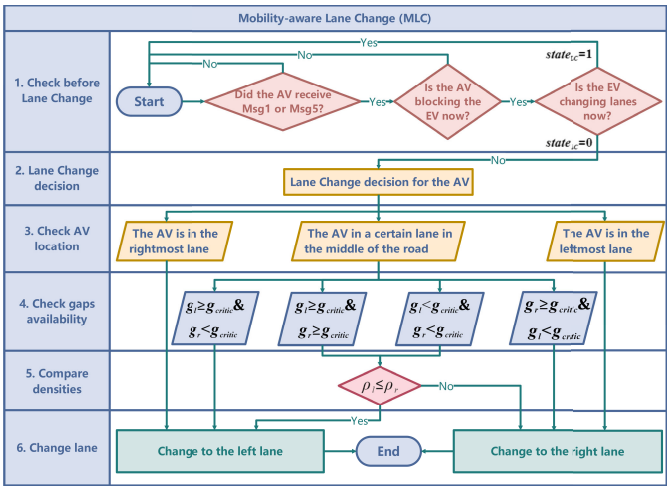

Fig. 5. Decision tree of the MLC algorithm.

C. QoS of V2X Messaging

Both the EV and AV lane-changing algorithms proposed in this paper are strongly rely on the QoS of V2X messaging, and poor PRR and EM delay performances will make these algorithms ineffective or even highly risky. However, the QoS of V2X messaging depends on a number of correlating network parameters, such as the communication range R, carrier sensing parameter σ, packet rate λ, and contention window W_c. For instance, a lower packet rate increases the channel load, thus EM delay would drastically increase; a higher packet rate reduce signal strength and limit the transmission range making the packets unreachable to distant vehicles [27]. Taking these factors into consideration, we propose an adaptive parameter adjustment mechanism in QoS of V2X messaging. In this algorithm, network parameters are determined based on the current vehicular density, which is presented as follows.

1) Vehicular Density and Average Speed: Vehicular density is inversely proportional to average speed as specified in [28]. We assume that OBUs and RSUs know the current average speed v_{arg}, the maximum allowed speed v_{max}, the current vehicular density ρ, and the maximum and minimum vehicular density ρ_{max} and ρ_{min} on the road. We introduce a vehicular density level parameter β, the exact equation of which is given in Algorithm 1. We empirically divide β into 10 classes from 0 to 9, where less than or equal to 30% ($\beta \leq 3$) is low density, 30% to 70% ($3 < \beta \leq 7$) is medium density, and more than 70% ($\beta > 7$) is high density.

2) Communication Range and Carrier Sensing Range: It can be known from [17] that reducing the communication range (i.e., radius) can effectively improve the QoS in congested traffic conditions. Therefore, we propose to adjust the communication range within the maximum and minimum range (R_{max} and R_{min}) according to the vehicular density in QoS-V2X. The carrier-sensing range (i.e., carrier-sensing radius) R_c is inversely proportional to the carrier sensing parameter σ. In case of high vehicular density, reducing the carrier-sensing range can reduce the waiting time for each vehicle to transmit its status message. Although that may create hidden terminal problems and hence packet collision, Algorithm 1 solves this issue by adjusting the communication range simultaneously, in which we configure σ to be 1, 0.5, and 0.25 for high density, medium density, and low density, respectively. The values of σ are chosen based on intensive simulations, as can be seen from the simulation results, such setting works well.

3) Packet Rate and Contention Window: Packet rate is the number of packets transmitted per second, it has been tuned lower to reduce channel load as vehicular density increases in the QoS-V2X protocol. Specifically, the packet rate λ is determined by the density level (i.e., $\lambda = 10 - \beta$). The contention window (CW) is the number of required backoff periods before a packet transmission. In QoS-V2X, we apply the curve fitting method in [29] to identify the optimal CW size that minimize the transmission delay with respect to the vehicular density. Furthermore, the size of CW (i.e., $W_c \in [8, 120]$) has been adjusted dynamically based on the EM delay T_{EM} to avoid packet collisions. When T_{EM} increases over 20% of the previous delay T'_{EM}, the size of CW will be increased to reduce contention.

The EV, AVs, and RSU execute the QoS-V2X every T second, where they update the vehicular density. The pseudocode of the QoS-V2X algorithm is shown in Algorithm 1. First, whenever T is 10 times the control channel interval, the EV calculates the vehicular density ρ according to the current average speed, thereby calculating the density level β. Then, according to β, the allocation of communication parameters (i.e., R, λ, σ, W_c) under high, medium, and low vehicular densities is decided.

Algorithm 1 QoS-V2X

Input: v, v_{min}, v_{max}, T_{EM}, T'_{EM}, F
Output: ρ, β, R, λ, σ, W_c

Initiate: $R \leftarrow R_{max}$; $\lambda \leftarrow 10$; $\sigma \leftarrow 0.5$; $W_c \leftarrow 8$
while $T == 10 \cdot CCI$ **do**
  $\rho = \frac{F}{v}$; $\rho_{min} = \frac{F}{v_{max}}$; $\rho_{max} = \frac{F}{v_{min}}$
  if $\rho \in [\rho_{min}, \rho_{max}]$ **then**
    $\beta = \lfloor \frac{\rho}{\rho_{max}} \cdot 10 \rfloor$ (β represents the density level)
    $\lambda = 10 - \beta$
    if $\beta > 7$ (high density) **then**
      $R = R_{min} + \beta \cdot \frac{R_{max} - R_{min}}{10}$; $\sigma = 1$ (i.e., $R_c = R$)
    else
      if $3 < \beta \leq 7$ (medium density) **then**
        $R = R_{min} + \beta \cdot \frac{R_{max} - R_{min}}{10}$; $\sigma = 0.5$ (i.e., $R_c \in [R, 2R]$);
      else
        (low density)
        $R = R_{max}$; $\sigma = 0.25$ (i.e., $R_c = 2R$)
      end if
    end if
    if $T_{EM} > 1.2T'_{EM}$ **then**
      $W_c = min(W_c + 16, 120)$
    end if
    EM $\leftarrow$ (Msg 1, Msg 2, Msg 3, Msg 4, Msg 5);
    Broadcast(EM)
  end if
end while

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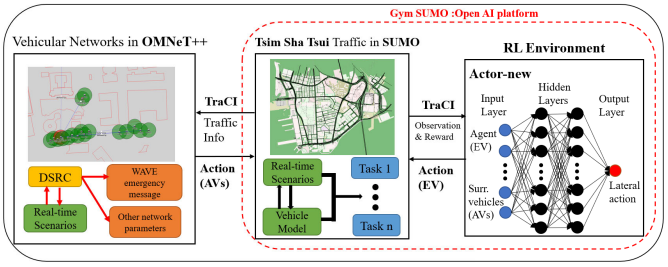

Fig. 6. Simulation Implementation.

V. Numerical Results and Performance Evaluation

The performance of the proposed algorithms are verified in this section by comparing numerical simulation results with multiple benchmarks. Fig. 6 depicts the overall architecture of the simulation, which is based on our previously proposed digital twin system for vehicular networks [30]. We employ SUMO as a microscopic traffic simulator with a realistic city model generated from open street map (OSM) [31] to simulate V2X and DRL lane changing at the same time. MLC is implemented using OMNeT++, and the mobility model is consistently input from SUMO in real-time via the traffic control interface (TraCI). Another TCP port is opened for the OpenAI Gym toolkit [32] to implement the DRL through TraCI for the SUMO server.

Since the simulation involves multiple software and they have built-in result extraction formulas, we further explain the process and source of result extraction. In each round of the simulation, the delay and packet reception rate (PRR) of V2X emergency messages are directly derived from OMNeT++, while the average speed of EV, travel time and lane change frequency of EV and AV are directly derived from SUMO. These results are collected and plotted in MATLAB. As for the DRL, the Keras API provides average cumulative reward data and plots, while recording the training time.

A. Simulation Parameters

In these experiments, we use the same initial parameters to compare different algorithms fairly. We chose Salisbury Road in Tsim Sha Tsui as the simulation road, which is a major road in Hong Kong's central business district (CBD) with vehicles running in a single direction on three lanes. The EV's start and end points are 2000 meters apart on the road, and the speed limit of Salisbury Road is 50 km/h. For car-following models, the cooperative adaptive cruise control (CACC) [33] is employed by AVs while the Krauss [34] is employed by the EV. IEEE 802.11p is used in the physical and MAC layers, and the wireless access in the vehicular environment (WAVE) [35] is considered in the network, transport, and application layers. In the simulation, we set up a normal traffic light system (TLS) according to OSM data, and the EV is allowed to run red lights. The rest of the parameters are presented in Table II. Note that the proposed algorithms in this paper are not restricted to the DSRC protocol stack and the WAVE, which are merely considered in the simulation to demonstrate the feasibility of the proposed algorithm only.

TABLE II
Parameters in Simulation

Parameter	Value	Unit
Simulation duration	3600.00	s
Number of lanes N_L	3	N/A
Straight-line distance	2000	meter
Mobility model	EV, AV, arbitrary starting points of the road	N/A
Number of vehicles in each lane α_i	α_i for the i-th lane; $\alpha = \sum_{i=0}^{N_L} \alpha_i$	vehicles/s
Limited speed on the road v_{min}, v_{max}	v_{min}=0, v_{max}=100	km/h
Vehicle density d_c	0.05, 0.10, 0.15, 0.20, 0.25	veh/m
Physical layer	IEEE 802.11p	N/A
Transport, network, and application layer	WAVE (i.e., IEEE1609)	N/A
Modulation and data rate	BPSK, 3	Mbps
Emergency Message and header sizes	512, 64	bytes
Signal fading model	Nakagami model	N/A
Initial transmission power P_t	20	mW
Initial communication range R for EV and AV	50	meter
Communication range for RSU R_{rsu}	200	meter
Initial carrier sense parameter ρ	0.5	N/A
Initial packet sending rate λ	10	packets/s
Initial contention window W_c	8	N/A
Control channel interval (CCI)	100	μs
slot time ϱ and propagation delay δ	13, 1	μs

The Actor-new, Actor-old, and Critic neural networks have the same setup, all implemented a five-layer fully connected network: the input layer, three hidden layers, and the output layer. Each layer has 29, 64, 128, 64, and 1 neurons, respectively. We use the ReLu function $f_{ReLu}(\omega) = max(0, \omega)$ for the hidden layers. The number of epochs is set to 1,000 and an epoch ends when 128 steps pass. The Huber loss and Adam optimizer are employed, with a 0.001 initial learning rate and a batch size of 32. The above values are obtained based on a trial-and-error method to avoid overfitting and underfitting during training. For training and testing, we use Keras, a deep-learning framework based on TensorFlow, and NVIDIA CUDA for the GPU back-end to reduce time. Three training runs were conducted to attain the best performance for the LC models. We trained it with the default parameters in the first run to establish a baseline; in the second run, we reduced the weight of the safety reward W_S, increased the weight of the comfort reward W_C, and decreased the weight of the speed reward W_V; and in the third run, we increased the learning rate to 0.01.

B. EM Delay Performance

Fig. 7 compares the emergency message delay of the proposed DRL-MLC with other benchmarks. Note that MLC is the second stage of the algorithm and DRL-MLC is the final algorithm. The difference between DSRC and MLC is that DSRC does not assign network parameters based on mobility. It can be calculated from the result that at relatively low density (i.e., 0.1 veh/m), the EM delay of DRL-MLC is

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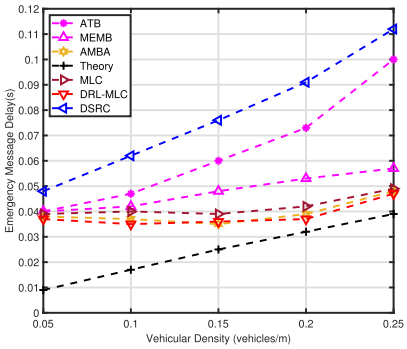

Fig. 7. Emergency message delay.

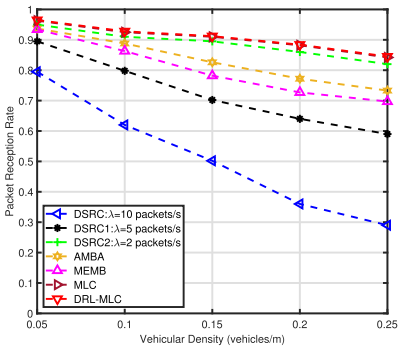

Fig. 8. Packet reception rate.

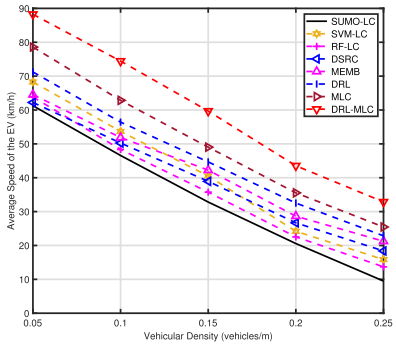

Fig. 9. Average speed of the EV.

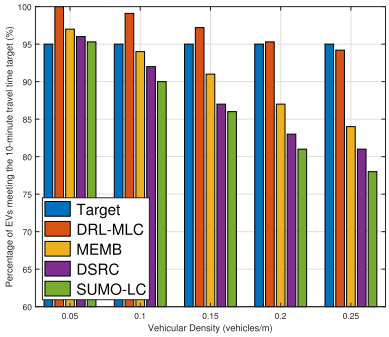

Fig. 10. Percentage of EVs meeting the 10-minute travel time target.

43.5%, 25.5%, 16.6%, 12.5%, and 5.4% lower than that of the DSRC, ATB, MEMB, MLC, and AMBA, respectively, but it is still far from (i.e., 51.4% higher) the theoretical value derived from (7). However, at a relatively high density (i.e., 0.2 veh/m), the EM delay of the DRL-MLC is 59.3%, 49.3%, 30.1%, 11.9%, and 5.1% lower than that of the DSRC, ATB, MEMB, MLC, and AMBA, respectively, and it is close to (i.e., 13.5% higher) the theoretical value. Therefore, the DRL-MLC algorithm effectively decreases message delay and is more suitable for scenarios with high vehicular density.

C. Packet Reception Rate Performance

Fig. 8 shows the PRR performance of the proposed algorithm and other benchmarks. DSRC, DSRC1, and DSRC2 are emergency message services that do not apply the mobility-aware algorithm and use varying packet sending rates λ. We can observe from the DSRC, DSRC1, and DSRC2 curves that PRR is sensitive to the packet sending rate, and as λ increases, PRR reduces dramatically. The DRL-MLC algorithm, on the other hand, is completely adapted to λ, and the corresponding PRR at relatively medium density (i.e., 0.15) is 41%, 21%, 1.7%, 8.6%, 13%, and 0.3% greater than the DSRC, DSRC1, DSRC2, AMBA, MEMB, and MLC respectively.

D. Average Speed Performance

The average EV speed of the proposed algorithm and benchmarks is shown in Fig. 9. For a fair comparison, we use the same network and traffic parameters to implement the simulation for each algorithm. Note that DRL represents the first stage of the algorithm. SUMO-LC serves as the baseline, in which EV and AV have the same priority for lane change. The average speed of the EV of DRL-MLC is 81.8%, 66.9%, 53.2%, 47.3%, 41.3%, 33.6%, and 21.7% greater than that of the SUMO-LC, RF-LC, DSRC, SVM-LC, MEMB, DRL, and MLC respectively, at medium vehicular density (i.e., 0.15 veh/m). The average speed of MLC is 9.7% greater than DRL at medium density, we can see that the impact of V2X communication in lane changing is considerable.

E. Travel Time and Lane Change Frequency

To verify the performance of the proposed algorithm from a macroscopic transportation perspective, we scale up the simulations using the entire region of Kowloon, Hong Kong, and the simulation times are 1,000 for each density to evaluate the overall performance of the proposed algorithm. The EV's starting and ending locations are stochastic implementations, the EV's travel distance is between 2,000 and 20,000 meters, and the rest of the parameters are identical to those in Table II. As defined in [36], the Hong Kong government has established a target for emergency ambulance vehicles of a ten-minute travel time performance of 95 percent for emergency calls. In 1996, however, just 89.5 % of emergency ambulances arrived within 10 minutes in Hong Kong. Only our proposed DRL-MLC algorithm, as shown in Fig. 10, achieves the 95% target, with an average of 97.1%. The percentage of EVs with a travel time less than 10 minutes in the DRL-MLC is 2.2%, 6.2%, 10.2%, and 11.2% percent higher than the Target, MEMB, DSRC, and SUMO-LC at relatively medium density. When the vehicular density increases, the performance of DRL-MLC gradually decreases and is lower than the target at extremely

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TABLE III
TRAVEL TIME COMPARISON

Algorithm	EV's average travel time (s)	AV's average travel time (s)	EV's lane change frequency	AV's lane change frequency
DRL-MLC	397.23	689.32	2.67	0.128
MLC	442.27	686.08	1.23	0.181
DRL	499.06	659.15	3.12	0.0826
MEMB	528.27	638.41	2.34	0.0815
SVM-LC	561.60	623.57	2.25	0.0822
RF-LC	603.39	612.65	2.01	0.0817
SUMO-LC	617.43	607.24	1.18	0.0821

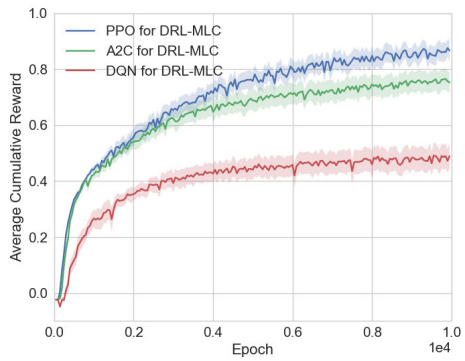

Fig. 11. Average cumulative reward per epoch.

high density (0.25 veh/m). This is because the increase in density leads to congested roads, and the number of lane changes that the DRL-MLC can perform decreases.

In addition, we compare the average travel time of EV and AV for each algorithm as shown in Table III. We can see that there is a trade-off between the EV's average travel time and the AV's average travel time. The average EV travel time of DRL-MLC is 10.1%, 20.4%, 24.8%, 29.2%, 34.1%, and 35.6% lower than that of MLC, DRL, MEMB, SVM-LC, RF-LC, and SUMO-LC. Meanwhile, the average AV travel time of DRL-MLC is also the highest (e.g., 13.1% high than that of SUMO-LC). This result reveals that DRL-MLC reduces EV's travel time by 35%, by sacrificing 13% of AV's travel time performance compared to SUMO-LC. We also introduced a metric, lane change frequency, which represents the number of lane changes per vehicle per kilometer. We can see that DRL-MLC achieves a good balance between EV and AV's lane change frequencies compared to other benchmarks.

F. Impact of RL Algorithms

To evaluate the performance of the proposed PPO model, we implement distinct RL algorithms for comparison, which are A2C and DQN. According to the experimental observation, performance differences of RL algorithms is more significant under low vehicular density. Therefore, we set the vehicular density of 0.05, and the rest of the simulation environment is exactly the same as the previous sections.

Fig. 11 shows that the rewards of PPO, A2C, and DQN all increase gradually with epoch and converge at 600,000 epochs. The total reward of PPO after convergence is about 11% and 66% higher than that of A2C and DQN, respectively.

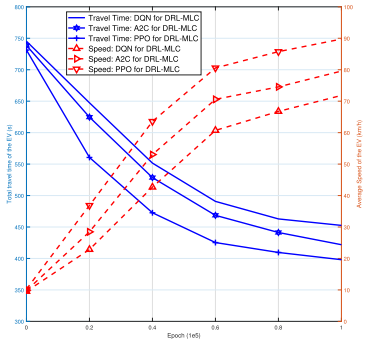

Fig. 12. EV traveling performances.

Moreover, as shown in Fig. 12, the EV travel time of PPO is lower than A2C and DQN by 20%, 33%, and the average speed is higher than A2C and DQN by 12.5%, 28.6%. It can be seen that DQN has the worst performance, because it is a value-based (off-policy) RL algorithm, which can converge and improve its results overtime but the final results are not outstanding. Both PPO and A2C can achieve better results by using an algorithm that learns the policy to follow, and A2C uses a pair of actor-to-critic networks for policy learning and weight update respectively. On the basis of A2C, PPO uses clipped surrogate objective to prevent the agent from becoming too greedy or too cautious in decision making of each action.

G. Time Complexity Analysis

In this section, we analyze the time complexity of the algorithms proposed in this paper. We compare the training and execution time complexities and training time of distinct LC algorithms, and rank them from low to high training time. Note that time complexity refers to the time in the worst case, and execution time refers to the duration from algorithm execution to LC decision.

As shown in Table IV, the SUMO-LC algorithm has the lowest execution time complexity ($O(n_{ru})$) and does not require training of the LC model. Note that n_{ru} is the number of the predefined rules and conditions in SUMO-LC. The DRL-MLC algorithms based on PPO and A2C use policy iteration to evaluates a policy and then use these values to improve that policy. Therefore, the training time complexity is the calculation of the value function and its policy update process, which is $O(|S|^2)$. The execution time complexity is the calculation of the value function to select the optimal policy, which is $O(|S|)$. The DRL-MLC algorithm based on DQN employs value iteration, which looks ahead one step and considers all possible actions at each iteration to find the maximum. Therefore, the number of actions $|A|$ must be taken into account in the time complexity. Noted that $|S|$ and $|A|$ represents the number of states and actions in DRL-MLC. In addition, the training and execution time complexities of RF-LC and SVM-LC are $O(n_t \log(n_t) n_{fe} n_{tr})$, $O(n_t^2 n_{fe} + n_t^3)$ and $O(n_{fe} n_{tr})$, $O(n_{fe} n_{sv})$, respectively, where n_t, n_{fe}, and n_{tr} denote the number of training samples, features, and trees. n_{sv} represents the number of support vectors.

In general, SUMO-LC has the lowest training time and execution time complexity, followed by the DRL-MLC based

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TABLE IV
TIME COMPLEXITY COMPARISON

Algorithm	Training time complexity	Execution time complexity	Training time (s)
SUMO-LC	N/A	$O(n_{ru})$	0
DRL-MLC(PPO)	$O(\|S\|^2)$	$O(\|S\|)$	20565.4
DRL-MLC(A2C)	$O(\|S\|^2)$	$O(\|S\|)$	25663.8
DRL-MLC(DQN)	$O(\|S\|^2\|A\|)$	$O(\|S\|\|A\|)$	55620.8
RF-LC	$O(n_t \log(n_t) n_{fe} n_{tr})$	$O(n_{fe} n_{tr})$	61080.5
SVM-LC	$O(n_t^2 n_{fe} + n_t^3)$	$O(n_{fe} n_{sv})$	80544.6

TABLE V
PERFORMANCE UNDER DIFFERENT SCENARIOS AND TRAFFIC DEMANDS

Scenario	Traffic demand	EV's travel time (s)			
		DRL-MLC	SUMO-LC	SVM-LC	RF-LC
Highway	Low	149.7	149.8	149.7	149.9
	Medium	212.8	277.4	258.2	265.3
	High	342.6	389.6	362.8	372.4
Road	Low	368.4	369.1	368.4	368.8
	Medium	549.2	623.3	586.2	597.8
	High	830.2	863.5	848.2	855.6

on PPO. In case of RTX 3070 GPU with CUDA, the training time of PPO is lower than that of A2C and DQN because it does less policy updates. When sampling a batch from the replay buffer and performing policy or gradient updates, if the batch size is large, then the time complexity will be very high. In experiments, we set PPO to do a policy update every two steps, A2C to do a policy update every step, and DQN to do a gradient update every step. The training time of RF-LC and SVM-LC algorithms is very high because of their high training time complexities. The training time can be shortened with fewer trees and support vectors, but the accuracy LC decision will suffer as a result.

H. Performance Under Different Scenarios and Traffic Demands

To further explore the suitable scenarios of the DRL-MLC algorithm, we designed experiments with different scenarios and traffic demands as shown in Table V. The scenario is set as a 3-lane highway or urban road with a length of 5 km, and the speed limit is 100 or 50 km/h, respectively. In addition, traffic demands are defined as low, medium and high demand, and their traffic flows are 500, 1500, 2500 veh/h, respectively. It can be seen from Table V that the EV's travel time of DRL-MLC is very close to that of other algorithms under the low-demand traffic of the highway and urban road. This is because the EV hardly make lane changes in the extremely sparse traffic. However, the computational complexity of DRL-MLC is much higher than that of SUMO-LC. Therefore, the low-demand scenario is more suitable for SUMO-LC, which has a sufficiently good performance and the lowest complexity. For medium and high-demand highway and urban road scenarios, DRL-MLC outperforms other algorithms, especially for medium demand cases. This is because the times of lane changing increases and the effect of the DRL-MLC algorithm is most pronounced. High-demand traffic generates a certain degree of congestion, which weaken the effect of DRL-MLC and increase the EV's travel time. Furthermore, because to the higher highway speed limits, each algorithm performs better under the highway than it does under the urban road.

VI. CONCLUSION

This paper exploits V2X communication technology and lane change control to minimize the travel time of emergency vehicles (EV). Specifically, we propose a connected vehicular environment and implement the DRL to allow the EV to change lanes at the right place and time, so as to identify the shortest trajectory. To further reduce the EV's travel time, we propose the MLC to make the blocked AV give way. In addition, the QoS-V2X is proposed to fine-tune the network parameters in real time for reliable V2X messaging. The results reveal that the DRL-MLC algorithm is more suitable for medium and high traffic flow scenarios, and its performance outperforms other lane change algorithms, such as SUMO-LC and SVM-LC. The proposed algorithm is practically feasible in real-world scenarios after solving the implementation challenges in the following three aspects.

1) High-level autonomous driving is required. DRL-MLC has complete vehicle control to make lane changes, therefore, it requires at least a level-4 autonomous driving defined by the society of automotive engineers (SAE), which means that vehicles are completely controlled by agents, and human override is only an option.
2) Real-time sensor data interaction is required. The agent and OBU use the data obtained by the sensors, calculates or combines them into states or encapsulates them into V2X messages. Such data interaction processes take time that causes delay in lane change execution. Therefore, DRL-MLC requires sensor data interaction as close to real-time as possible.
3) Reliable distributed control of vehicular information is required. The V2X messaging protocol is a kind of distributed control, which means that vehicular information is stored and exchanged individually at each vehicle node and RSU, rather than being uploaded to the central server for operation. When a node fails, we still need to keep the V2X messaging protocols functioning. Therefore, reliable distributed control is required to prevent protocol crashes.

Furthermore, we summarize the limitations of the proposed algorithm and framework, which paves the way for our future work. First, this work uses the Nakagami signal fading model to simulate the propagation, which does not consider the reflection, refraction and blocking effects of urban buildings on wireless signals. This will cause V2X messages transmission failure in real-world scenarios, thus degrading the performance of DRL-MLC. Second, this work assumes level-4 fully autonomous driving, which means that all vehicles obey DRL-MLC to perform lane changes. However, it is difficult to achieve in the real world in the short term since it is more likely that human-driven vehicles and autonomous vehicles will coexist. The performance of DRL-MLC will be influenced by human-driven vehicles that refuse to obey the algorithm, therefore, different proportions of human-driven vehicles should be considered in the future. For future work, we will enhance the vehicular signal propagation model, and take into account the urban building effect. Ray tracing techniques and

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machine learning will be incorporated into the modeling. In addition, in terms of connected autonomous driving, we will study the reaction of human-driven vehicles to V2X messages, and consider the interaction between human-driven vehicles and autonomous vehicles in the algorithm.

Finally, we provide insights into the potential impact of the proposed algorithm on the overall flow. Typically, there is a bottleneck here, the cost of improving EV efficiency is sacrificing the efficiency of non-emergency vehicles. The design of DRL-MLC comprehensively considers the efficiency of the EV and non-emergency vehicles, as well as the trade-off between them. While the blocking AV gives way to the EV, DRL-MLC considers the impact of the AV lane change on the overall traffic flow to avoid a certain lane from being too congested. In general, the proposed algorithm and framework in this groundbreaking work will open up a wealth of new research options for future connected autonomous driving.

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Cao Ding received the B.Eng. degree in automation from the Beijing University of Chemical Technology, Beijing, China, and the M.Sc. degree in electronic and information engineering from The Hong Kong Polytechnic University, Hong Kong, where he is currently pursuing the Ph.D. degree. His research interests include vehicular networks and intelligent transport systems (ITS), specifically in vehicular ad hoc network (VANET). His research focuses most on the digital twin of vehicular networks and intelligent transport systems.

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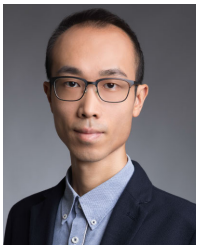

Ivan Wang-Hei Ho (Senior Member, IEEE) received the B.Eng. and M.Phil. degrees in information engineering from The Chinese University of Hong Kong, Hong Kong, in 2004 and 2006, respectively, and the Ph.D. degree in electrical and electronic engineering from Imperial College London, London, U.K., in 2010. He was a Research Intern with the IBM Thomas J. Watson Research Center, Hawthorne, NY, USA; and a Post-Doctoral Research Associate with the System Engineering Initiative, Imperial College London. In 2010, he has cofounded P2 Mobile Technologies Ltd., where he was the Chief Research and Development Engineer. He is currently an Associate Professor with the Department of Electrical and Electronic Engineering, The Hong Kong Polytechnic University, Hong Kong. His research interests include wireless communications and networking, specifically in vehicular networks, intelligent transportation systems (ITS), and the Internet of Things (IoT). He primarily invented the MeshRanger series wireless mesh embedded systems, which received the Silver Award in Best Ubiquitous Networking at the Hong Kong ICT Awards in 2012. His work on indoor positioning and the IoT also received the Gold Medal at the International Trade Fair Ideas and Inventions New Products (iENA) in Germany in 2019 and the Gold Medal with the Organizer's Choice Award in the International Invention Innovation Competition in Canada (iCAN) in 2020. He was the TPC Co-Chair of the PERSIST-IoT Workshop in conjunction with ACM MobiHoc 2019 and IEEE INFOCOM 2020. He is currently an Associate Editor of IEEE ACCESS and IEEE TRANSACTIONS ON CIRCUIT AND SYSTEMS—II: EXPRESS BRIEFS.

Edward Chung received the bachelor's and Ph.D. degrees from Monash University. He is a Professor of intelligent transport systems with The Hong Kong Polytechnic University (PolyU). Prior to his appointment at PolyU, he was a Professor and the Director of the Smart Transport Research Centre, Queensland University of Technology. He has many years of experience as an experienced academician and a researcher, working both domestically and internationally. His research interests include connected autonomous vehicles, artificial intelligence, traffic management, and transport modeling.

Tingting Fan received the B.Eng. degree in electrical engineering and automation from Jinan University, Guangzhou, China, in 2017, and the M.Sc. degree in electronic and information engineering from The Hong Kong Polytechnic University, Hong Kong, in 2022. She is currently a Research Assistant with the Department of Electrical Engineering, The Hong Kong Polytechnic University. Her research interests include intelligent transportation systems (ITS), vehicular networks, and reinforcement learning.

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